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Practices in Bio-design: Design Research Through Interdisciplinary Collaboration

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Chapter 4

Practices in Bio-design: Design Research Through Interdisciplinary Collaboration



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Abstract Within the emerging field of bio-design, there is a growing need for interdisciplinary practice and collaboration that enables design with living organisms. To facilitate such work, designers often venture outside of the limits of their own design field and engage with methods, knowledge and language of other disciplines. Bio-design collaborations range from fundamental studies to applied research that incorporates existing knowledge and ideas. The examples presented in this paper are part of the author's doctoral research in architecture by creative practice that include a fundamental study of photosynthetic bio-composite materials, an applied materials study through the use of fabrication with living organisms and an eco-philosophical graphic novel based on conversations with scientists. The outlined studies demonstrate the adoption of laboratory methods within design practice, the integration of existing scientific work within design and the organic exchange of ideas through alternative methods of visual communication. The paper explores the opportunities and limitations of such practices and highlights their importance within the field of bio-design. The presented studies help define a spectrum of interdisciplinary collaboration in relationship to discipline specific skills utilized depending on design intent and outcome. This paper seeks to highlight the benefits of knowledge and skills transfer through bio-design collaboration, offering a range of possibilities using different mediums and modes of communication between practitioners of the science and design fields.

4.1 Introduction

There is a long history of collaborations between designers and scientists that are at the basis of innovation and social change including the translation of scientific principals into commercial products, early stage prototyping and education and public engagement [1]. These types of collaboration have grown in importance with the

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emergence of bio-design [2]. In this hybrid field, the role of the designer is reframed to encompass practices that were previously considered outside of the scope of design, where design outcomes embrace scientific principles or designers participate in the development of biotechnology. Interdisciplinary collaboration takes on a central role within bio-design as there is an innate exchange of knowledge, skills and philosophical perspectives that takes place when working with living organisms. By working closely with other practitioners, architects are able to bring living organisms into a manmade realm either by integrating them into our everyday surroundings, or by bridging the gap between scientific practice and the rest of society [3]. This mode of design through the exchange of ideas between different fields was defined as *information design* by Andersson and Pettersson [4]. In the field of bio-design, this typically occurs through collaboration with scientists, industry or other bio-designers.

Despite the growing focus on the field, much remains to be said about how these collaborations can occur and in what forms. This paper reflects upon the similarities within the scientific research and design processes and the informal knowledge exchange within interdisciplinary collaboration and demonstrates how these factors influence three distinct forms of bio-design work. The highlighted examples serve to illustrate three types of bio-design research that are structured to achieve various objectives including fundamental investigation through scientific practice for the developing of a living bio-material, adoption of scientific principles to generate a living materials installation, as well as a design-led exploration of language to propose design solutions through the use of storytelling and drawing. The examples presented in this paper are part of the author's doctoral research by creative practice and tell the story of the presented collaborations from the perspective of an architect collaborating on interdisciplinary bio-design projects. The collaborations vary in duration as well as method and range across a wide spectrum of collaborative exchange, from the adoption of science into a design practice, to long-term laboratory experimentation where designers learn, adopt and adapt scientific methods so as to understand processes and work towards solutions that can be scaled and employed outside of a laboratory setting.

4.2 Scientific and Design Processes

Science has historically assumed a positivist approach to knowledge, one that is limited to the experiences of the senses and that dismisses metaphysical explanations [4]. In more recent history, science has also allowed for ideas from the hermeneutic tradition that has a more expansive view of knowledge and that accepts methods such as acquiring meaning through doing [5] or making as is the case with design, providing a common ground for collaboration. Furthermore, scientific and design research both engage in a creative process that can be divided into four stages as defined by Graham Wallas, which include preparation, incubation, illumination and verification [6]. However, in both cases whilst these stages occur, they do not necessarily unfold as a linear process as there is an element of uncertainty present within

both practices. Due to this uncertainty the bio-design process, although differing from scientific practices, it is in a state of flux where variables are gradually revealed and initial planning presents challenges, as the process must accommodate changes and evolution of ideas during the making process. These stages of investigation may take on the form of laboratory testing, prototyping or drawing as will be demonstrated in the examples. These similarities across disciplines serve as a common ground for collaborations between scientists and designers and are the focus of existing research that this paper builds upon, such as the work of Rust [7, 8] and Polanyi [9] as well as architectural bio-design research undertaken by architects and scholars such as Imhof and Gruber [10] and Armstrong [11] who engage in interdisciplinary research through practice. The examples in this paper expand upon such practices and demonstrate the types of knowledge transfer from the scientific to the design field identifying the adoption of tools and knowledge necessary for the three different types of design objectives, ranging from the furthering of knowledge with future application in mind, the scaled testing of novel biotechnology and engaging within scientific ethics through the use of design tools.

4.3 Casual Knowledge Exchanges in Interdisciplinary Collaboration

Within the examples presented in this paper, there is a level of informal exchange that precipitates learning and testing of ideas in each project. This paper advocates the importance of such informality within the field of bio-design and proposes that it is an essential tool for research through practice in an interdisciplinary setting. The paper seeks to demonstrate the types of collaboration that are available and appropriate depending on the type of bio-design practice and objective as well as to highlight the benefits and limitations of these kinds of practice.

Knowledge exchanges typically occur through conversation during the prolonged time spent in the laboratory by cultivating close relationships with scientists. Such input accelerates the learning process through the demonstration of methods, translation of scientific jargon and explanation of fundamental principles [12]. This type of learning, as observed by Sawa [13], is very much reliant on informal social interaction as part of a natural bonding process facilitated by proximity and frequency of encounters within the laboratory. The casual learning of methods and practices is a process that occurs over a long period and that cannot be facilitated through sporadic observation or formal interviews and discussion as in those cases, each participant is locked within the limits of their own professional reference point. In these instances, there is a danger of oversimplifying the intricacies of the other field through the overuse of accessible analogies necessary for communicating specialized knowledge to an outsider. It is not to say that such analogies are of little value as they enable the translation of ideas from one form of practice to another often to serve a different purpose. For example, narrative development, as shown within the

final example, captures the conversations that take place within scientific settings and communication of ideas by scientists, in a language foreign to laboratory culture. The graphic novel helps identify an attitude towards the work and aids an understanding of the world gained from working within science. Designers and architects translate information to create scenarios of future social systems and propose potential manifestations of such technology into scalable global strategies that do not abide by the limitations of current levels of scientific progress.

4.4 Adoption of Laboratory Methods: Material Development

The exclusive space that constitutes the laboratory can be intimidating and inaccessible both within the legal structure that governs it, as well as the highly specialized spaces that are hidden behind doors full of paraphernalia that is not referenced in everyday life [14]. However, with the growing dialogue between designers and scientists, bio-designers are beginning to embrace laboratory testing within their work that allows them to participate in the early stages of material development that permits participation in the shaping of a new type of living building fabric as demonstrated in the following project (Fig. 4.1).

An example of such work is testing of photosynthetic ceramic bio-composite material experiments conducted as part of the author's doctoral research by creative practice in architecture. In this instance, the work was undertaken by an architect following, modifying and subverting existing scientific methods [15]. The work was preceded by a period of shadowing of science researchers as well as the building of basic scientific knowledge and understanding through a theoretical survey and informal conversation. This preliminary work is fundamental to the ability of a

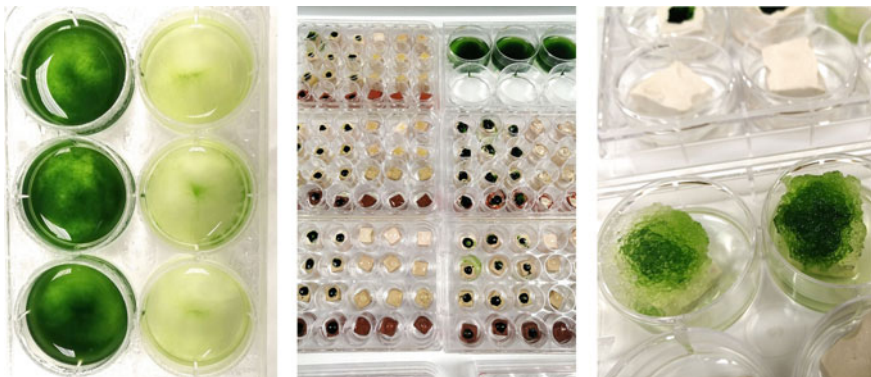


Fig. 4.1 Microalgae (*Chlorella vulgaris*) grown on various types of ceramic substrates as part of a laboratory experiment (work by author, in collaboration with Dr. Gary Caldwell and Pichaya In-na)

design practitioner to operate within a laboratory setting and is dependent upon a strong interdisciplinary partnership. By being able to adopt scientific methods of testing, the research focus was directed at material substrates and organisms that would be suitable for application within architectural design projects [16]. Furthermore, the types of tests were directed at enabling the organism to function outside of a controlled laboratory setting, an aspect often overlooked during fundamental testing as the scientific emphasis at these early stages of experimentation is on obtaining repeatable data [17]. Controlled experiments are predominately studied within laboratory research as they provide a consistency of results; however, they are not reflective of real-life environments that are of interest to designers. Whilst scientists in the laboratory may be undertaking similar type of investigation, there is a clear distinction between the roles of the scientist whose work is centred around a deep understanding of biological processes through the use of a wide range of data. This is in contrast to the work of the designer whose mission is to develop solutions with application as the end goal. The level of knowledge and understanding of the designer is likely to be limited to the particular area of investigation and is focused on data that determines tangible performance. Therefore, there is a reliance on the scientific counterparts to provide guidance and advice as experimentation unfolds and results bring about new questions. From a design perspective, the value of engaging in laboratory experimentation lies in the performance of the experiments as an act of making as well as in the data acquired. Although the processes may be identical, the objectives of the design practitioner are influenced by the tacit knowledge acquired through engagement in another discipline's practices and work environment. This knowledge serves to define future experimentation and guides the direction of research.

4.5 Integrating Science Within Design Practice: Yggdrasil

Bio-design has been adopted by many design practitioners and these works often employ or build upon existing scientific research that they integrate into design applications. Practitioners have utilized biological processes and living organisms within a variety of application including fashion [18], interior design [19] and building materials [20, 21]. Such works often emerge from collaborations between bio-designers that can harness each other's acquired scientific knowledge and understanding of biological principles to inform a hybrid practice as shown in Fig. 4.2.

Yggdrasil, a bio-design exhibit presented at the OpenCell Exhibition part of London Design Festival, 2019 [22] featured a range of experimental bio-design pieces demonstrating early testing of biological materials. The exhibit was created through a close collaboration between the author and other bio-designers with a background in architecture, where material research conducted as part of laboratory practice was used to create a living materials installation. The collection of work speculated as to the potential of biological matter to be used as building material and the fragility of the organisms used in such novel technologies, introducing an ethical component to



Fig. 4.2 Yggdrasil, bio-design exhibit at OpenCell, London Design Festival, 2019 (Thora Arnardottir, Assia Stefanova, Dilan Ozkan, Sunbin Lee)

material practice. The work highlighted the ability of these new approaches to create relationships within our immediate environment establishing connections between different actors [23], human and non-human, creating partnerships and exchanges. This type of design acts as a bridge between scientific and design practice, where design skills are more actively incorporated and the learning curve is less significant for designers as there is a greater emphasis on issues of scale, form and integration within the built environment that are integral to the field of architecture and are a familiar challenge encountered by designers. The type of knowledge exchange necessary for this type of bio-design work does not necessarily occur exclusively between scientists and designers. The source of knowledge can become practitioners and experts that do not necessarily feed into an institutional research framework, but utilize biological process within different artistic or commercial fields. Outside of the external sources of knowledge bio-designers have the ability to come together and bring their acquired expertise regarding organisms and methods, as was the case of *Yggdrasil*. The collaboration led to further fundamental research testing and use of laboratory protocols to investigate the compatibility of these hybridized solutions. Therefore, such collaborations can prompt creative thinking and early prototyping that demonstrates the scientific principals and aids public engagement at an early stage of the research as was the aim of the exhibit.

4.6 Casual Exchange for Narrative Creation: Ecological Graphic Novel

Many collaborations between scientists and artist occur through initiatives designed to make science accessible to the public. These collaborations are often based on

observation of scientific practice, techniques and conversations, between scientists and design practitioners. The designer in this case is entrusted with translating scientific practice into a media that is accessible to the public, whilst proposing further interpretation and application of scientific ideas, a practice that goes beyond literal communication. The media vary from employing scientific artefacts such as tissue culture as in the case of art works such as *Victimless Leather* [24] or creating proxies through digital simulation and fabrication to explain principles [25] or the use of various types of storytelling [26–29]. In an effort to propose wider integration as well as to address ethical issues, narrative creation becomes a means for testing of ideas on a wider scale; it helps identify social issues that are likely to affect implementation such as the work in Fig. 4.3.

A bio-design narrative example is the graphic novel, *Bios in Search of Zoe*, a work created as part of the author’s creative practice doctoral research in architecture. The work provides a glimpse into scientific practices and offers a critique of certain types of scientific investigation whilst proposing strategies for implementation of biological processes to create a new type of built environment. The story centres around the dualism between Bios and Zoe that originates from the two Greek terms for life. Bios embodying nature as controlled and beneficial to humankind and Zoe, representing nature as all-encompassing life, without an anthropocentric bias [30]. The work is based on conversations between scientists and architects and the existing research on ethics of working with microorganisms [31–33]. The graphic novel in this example is a method of research that positions bio-design practice into a broader social and political context as explored by social scientists, anthropologists and ecological thinkers [34–36]. Humanity’s disruption of natural networks provides the starting point of the narrative that as the story unfolds, places humanity within a complex multi-species entanglement that proposes an ethical approach to working with living organisms. The narrative serves to illustrate how biological solutions could feed into a broader

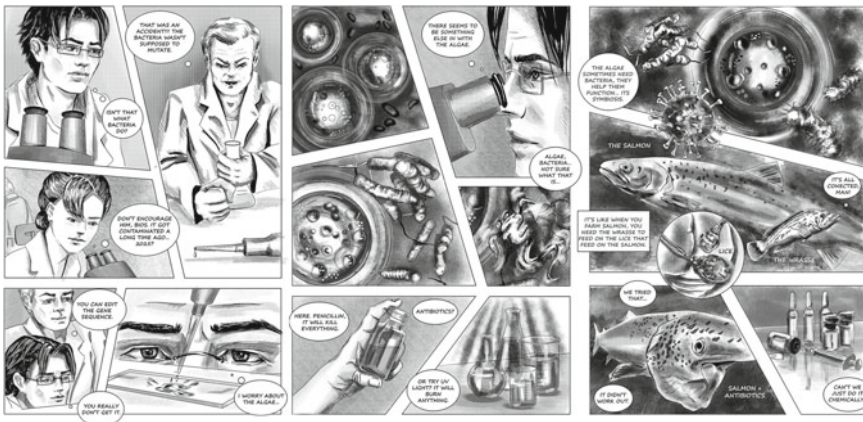


Fig. 4.3 Excerpts from the graphic novel “*Bios in Search of Zoe*”, work in progress as part of the author’s Ph.D. by creative practice research (work by Author)

global practice and the societal restructuring necessary to allow wide scale adoption of such technology. The use of fiction and drawing practice permits a critical examination of the implications of adopting emerging biotechnologies whilst reflecting on ethical issues of collaborating with living organisms. This type of synthesis of scientific practice serves to encourage engagement of non-scientific audience and allows the public to participate in scientific debates. This interpretation of conversations goes beyond the understanding of problems and the accurate representation of scientific facts. Designers take on the responsibility of interpreting tacit knowledge acquired during informal conversations, a process of *indwelling* or emersion within the discipline as defined by Michael Polany, a type of learning prevalent within both the science and design fields.

4.7 Discussion

The studies presented demonstrate three different points on the spectrum of bio-design collaboration. They identify the types of skills and knowledge exchange that could potentially occur, with practices situated closer to the social realm involving greater use of design tools whilst, practical performance-based objectives require a greater level of engagement with scientific methods as demonstrated in Fig. 4.4.

Use of scientific methods and laboratory practice within design development allows designers and architects to actively participate in the creation of biotechnology and guide the direction of the research with application in mind. In the field of architecture, this means that testing can be directed so that it takes into consideration conditions offered by the built environment, from the early inception of such research. This mode of working within a science setting presents challenges in terms of bridging a wide knowledge gap that designers must overcome in order to engage effectively in fundamental biological material development and testing. It highlights the reliance on willing and receptive scientific partners that could facilitate and benefit from designer input at an early stage of the research [37]. This type of

ENGAGING IN SCIENTIFIC PRACTICE	ADOPTING SCIENTIFIC RESEARCH	DESIGN REPRESENTATION
Scientific Skills		Design Skills
Scientific protocols, tools and practical knowledge are adopted, replicated, and used by design professionals within a science research setting.	Knowledge and practices from science and design disciplines are used within design solutions with users, application and social and economic considerations in mind.	Design skills and assessable language are employed to bridge the gap between science and the rest of society using design media, metaphors and proxies.
Material development, controlled testing, design of protocols	Product development, real world testing, practical application	Public engagement, social change and communication

Fig. 4.4 Spectrum of discipline-related skills against type of practice and their potential application

practice requires a greater level of understanding and the gradual building of foundational knowledge that enables the design requirements to guide the development of solutions.

In contrast, by adopting existing scientific research, the generated design work has the potential to bridge laboratory experimentation and scaled prototyping, situating the work closer to a traditional design practice. This type of testing differs from the commercial scaling of developed biotechnology and places a greater emphasis on bridging the gap between the exclusive practices adopted in early stage scientific investigation and the public. However, there is a danger that the performance of the living organisms could be overlooked in favour of design considerations through a lack of in-depth understanding of biological processes on the part of the bio-designer or the lack of testing equipment necessary to assess the health of the organism. In the example of *Yggdrasil*, there was a recognition that the installation provided a working demonstration of materials and was not a functional product and quantitative data could not be obtained. However, the role of *Yggdrasil* was to serve as a catalyst for further fundamental research made possible by engaging in both prototyping and laboratory testing simultaneously, which helped to identify future areas of exploration.

The issue of language becomes more problematic as different disciplines employ jargon and assign different levels of exactitude to language. This is particularly challenging when the design work aims to communicate principles and specialized methods to an audience that may not necessarily possess a foundational understanding of such ideas. Communication takes on less prescriptive role in graphic design, and in the case of *Bios in Search of Zoe*, it invites the reader to engage with issues that are presented as part of a broader discussion. As designers we use tools from design disciplines to interpret the information, propose solutions and anticipate plausible scenarios. Through the use of a drawing practice common within the field of architecture, bio-designers engage in an intuitive process that relinquishes control of the viewer's interpretation, which is in contrast to scientific processes of problem identification and theory modelling. The presented graphic novel builds upon an investigative drawing methods employed by other researchers within the field of architecture, examples include *Practiceopolis* and *Citizens of No Place* [38, 39]. In these instances, the architectural profession and design concepts are challenged through the construction of visual compositions that carry embedded information in an image format that influences the viewer on a subconscious level. Further exploration is necessary to determine the impact that such work has and the level of engagement and understanding it can facilitate within the field of bio-design.

4.8 Conclusion

The presented case studies feed into a spectrum of interdisciplinary collaborations typical of the emerging field of bio-design. The projects highlight the necessity for

such working partnerships and demonstrate bio-design research and testing strategies through a variety of working methods and different levels of engagement with scientific research. The studies have identified that language and communication are integral parts of bio-design collaboration and knowledge exchange. However, language also presents challenges in the differences in terminology which can cause misinterpretation or could create barriers that need to be overcome by designers when engaging with scientific knowledge. Designers are therefore able to postulate outcomes and draw on expertise outside of the scientific field to add a layer of critical reflection within bio-design research. The examples presented demonstrate work that encompasses fundamental and applied research in the field of bio-design that enforces the notion that the boundaries between the two are less clearly defined than commonly accepted. The co-creative process serves as a catalyst that prompts thinking regarding applied research during the fundamental testing phase or as a method of reflection upon the wider social impact of scientific practice.

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